

# Heterogeneous Transient MODFLOW 6 Benchmark

A strict test case for PINN and ML groundwater surrogates

Forward (simulation) and inverse (parameter estimation) problems

|                     |                                                                   |
|---------------------|-------------------------------------------------------------------|
| Generated           | 2026-08-06 00:35                                                  |
| Realisation seed    | 20260805                                                          |
| Grid                | 5 layers x 100 rows x 100 cols @ 100 m (10 x 10 km)               |
| Active cells        | 43,805 of 50,000                                                  |
| Time                | 360 stress periods x 2 h = 30 days, 1 time step each              |
| Packages            | DIS NPF ST0 IC HFB RIV GHB WEL RCHA SFR OBS OC                    |
| Solver              | IMS COMPLEX / BICGSTAB / DBD under-relaxation, Newton formulation |
| Run time            | 92.5 s                                                            |
| Budget discrepancy  | max incremental 0.0000 %, cumulative 0.0000 %                     |
| Fault head contrast | 9.4x the background neighbour-cell head gradient                  |
| Stream sinuosity    | 1.705 (163 SFR reaches, 139 RIV cells)                            |

# 1. Conceptual model

## PURPOSE

A deliberately hostile synthetic aquifer for benchmarking machine-learning surrogates. Every feature below exists to defeat a specific shortcut a surrogate might otherwise take: short-range heterogeneity defeats low-rank approximation, the fault barriers defeat smooth function representations, the meandering channels and the pinched-out layer defeat grid-aligned assumptions, and the 2-hour forcing defeats temporal subsampling.

## DOMAIN AND GEOMETRY

10 x 10 km, 100 m cells, five layers. The land surface falls from a northern plateau (mean 58 m) to a southern coastal plain (mean 50 m), carrying a kilometre-scale undulation (std 1.8 m) and hectometre-scale roughness (std 0.5 m). Two meandering watercourses incise it: the central stream cuts a valley 9 m deep and the tidal river a floodplain 6 m deep. Where the two overlap the deeper incision wins, which produces a proper confluence rather than a double-cut trench.

Recent Alluvium (Layer 1) fills that incised corridor, thickening downstream from 10 m to 18 m on the axis and tapering to nothing at the margins. Where the fill is thinner than 2 m the layer is pinched out (idomain = 0), so Layer 1's active domain is an irregular meander belt covering 38.0 % of the plan area - not a rectangle.

The four deeper contacts are undulating palaeo-surfaces about their mean elevations, so layer thickness - and therefore transmissivity and storativity - varies smoothly beneath the short-range K noise. A final pass enforces a minimum thickness on every layer so no active cell can be degenerate.

## STRATIGRAPHY

| Layer | Unit            | Top_m      | Base_m     | Thick_m   | Active | Kh_geomean | Convertible |
|-------|-----------------|------------|------------|-----------|--------|------------|-------------|
| 1     | Recent Alluvium | 40.6..57.2 | 23.1..55.2 | 2.0..17.7 | 3805   | 15         | yes         |

# 1. Conceptual model (continued)

|   |                     |              |                |             |       |      |     |
|---|---------------------|--------------|----------------|-------------|-------|------|-----|
| 2 | Older Alluvium      | 23.1..59.9   | 9.5..25.4      | 10.1..45.0  | 10000 | 5    | yes |
| 3 | Massive Carbonate 1 | 9.5..25.4    | -6.7..7.5      | 5.0..27.6   | 10000 | 25   | no  |
| 4 | Massive Carbonate 2 | -6.7..7.5    | -55.1..-35.6   | 32.4..55.8  | 10000 | 1    | no  |
| 5 | Fractured Granite   | -55.1..-35.6 | -155.5..-138.7 | 91.3..112.5 | 10000 | 0.05 | no  |

## HETEROGENEITY

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gstools exponential-covariance Gaussian random fields, correlation length 200 m = 2 cells,  $\text{var}(\ln K) = 2$ .  
Kh is log-normal about the per-layer geometric means;  $K_z = K_h / 10$ ; Ss is log-normal; Sy is the same  
correlated field squashed logistically onto (0.01, 0.38) because a volume fraction cannot be log-normal.  
Each layer uses an independent seed, so there is zero vertical correlation - deep K cannot be inferred  
from shallow K.

Layer 1 additionally carries a channel-facies trend: K is boosted by up to 2.5x within 500 m of the  
palaeo-channel axis, because coarse channel-lag and point-bar deposits sit there while overbank silts sit  
at the corridor margins. This gives the inverse problem a physically meaningful structure to recover  
rather than pure noise.

## FAULTS (HFB, ALL FIVE LAYERS)

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Fault 1 impermeable    hydchr = 1e-08 1/d  
          (r10,c10) -> (r40,c35)  
Fault 2 semi-permeable hydchr = 0.001 1/d  
          (r85,c99) -> (r65,c65)

Each oblique trace is discretised as a 4-connected staircase of shared cell faces, which is the only  
construct MODFLOW's HFB package accepts: diagonally adjacent cells do not share a face and cannot host a  
barrier.

## 2. Boundary conditions, stresses and numerics

### RIV - MEANDERING TIDAL RIVER

-----  
139 cells tracing a two-harmonic meander across the southern domain, placed in the uppermost active of Layers 1-2. Stage follows an M2 semi-diurnal tide: mean 40 m, amplitude 2.5 m, period 12.4 h, modulated by a 14.77-day spring-neap envelope. With 2-hour stress periods the tide is sampled only ~6.2 times per cycle - close to Nyquist, so a surrogate that coarsens time will alias it. Bed at 36 m, conductance 8,000 m<sup>2</sup>/d per cell.

### SFR - MEANDERING CENTRAL STREAM

-----  
163 reaches from the northern headwater to the confluence, sinuosity 1.705. The bed falls monotonically from 49.99 m to 39.72 m (a running minimum enforces this - a stream cannot flow uphill and SFR needs a strictly positive gradient). Width 15 m, Manning 0.03, bed K 0.5 m/d.

Head-reach inflow is baseflow (4,000 m<sup>3</sup>/d) plus three asymmetric flash floods peaking at 200,000 m<sup>3</sup>/d. Two of the floods follow storms; one does not, so stream inflow cannot be inferred from the rainfall signal. The downstream reach discharges out of the model at the confluence - MODFLOW cannot route a mover from SFR into RIV, because RIV is not an advanced package.

### GHB - EASTERN REGIONAL INFLOW

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Column 99, Layers 2-4, head 48 m. Conductance is computed cell-by-cell from the local heterogeneous Kh and the local layer thickness, so even the boundary flux inherits the geostatistical roughness and the undulating stratigraphy. Cells shared with the tidal river are skipped so the two head-dependent boundaries never fight over one cell.

### WEL - PRODUCTION WELLS

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8 wells in Layer 2 within the alluvial corridor. Diurnal schedule (night baseline 0.25, smooth daytime peak 06:00-18:00), a weekend reduction to 45 %, per-well phase offsets so the composite is not a single clean harmonic, and two hard step changes: W4 starts at day 12, W6 stops at day 20. Total abstraction swings between 434 and 7,049 m<sup>3</sup>/d.

## 2. Boundary conditions, stresses and numerics (continued)

### RCH - AREAL RECHARGE

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Base 0.0005 m/d with two intense 12-hour storms at 6-6.5 d and 20.5-21 d at 0.06 m/d (30 mm each). IRCH is set explicitly to the uppermost active layer: MODFLOW's array recharge does NOT fall through a pinched-out top layer, and omitting IRCH silently loses all recharge outside the meander belt.

### NUMERICS

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Newton formulation with under-relaxation; IMS COMPLEX complexity, BICGSTAB acceleration, Delta-Bar-Delta under-relaxation and backtracking. Layers 1-2 are convertible, 3-5 confined. Initial heads come from a steady-state spin-up under base forcing - base recharge and median baseflow, because using the series means would fold the storms and the floods into the initial condition and start the benchmark from an artificially wet aquifer.

### RESULT QUALITY

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|                                        |                                  |
|----------------------------------------|----------------------------------|
| Maximum incremental budget discrepancy | 0.0000 %                         |
| Final cumulative budget discrepancy    | 0.0000 %                         |
| Mean fault head-jump contrast          | 9.4x background                  |
| Water table above land surface         | 1.9 % of columns<br>(max 4.62 m) |

### CAVEAT ON THE PRESCRIBED PARAMETER SET

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The prescribed base recharge over 100 km<sup>2</sup> is large relative to what the prescribed transmissivities can route to the two outlets. In the small fault-dammed compartment behind the impermeable fault the water table therefore still reaches land surface, and because MODFLOW switches those cells to confined storage above the layer top, the storms produce head spikes of several metres there. That is a property of the parameter set, not a numerical failure - the budget closes to 0.00 %. Physically it is where springs and seepage faces would form. Lower Config.rch\_base or add a DRN seepage package at land surface if a strictly sub-surface water table is required everywhere.

### 3. Properties and diagnostics

HYDRAULIC PROPERTY STATISTICS (ACTIVE CELLS ONLY)

| layer | unit                | n_active | kh_geomean | kh_min    | kh_p05   | kh_p95 | kh_max | kh_log10_range | ss_mean   | sy_mean |
|-------|---------------------|----------|------------|-----------|----------|--------|--------|----------------|-----------|---------|
| 1     | Recent Alluvium     | 3805     | 19.32      | 0.1705    | 1.673    | 204.6  | 3289   | 4.285          | 0.0002389 | 0.2145  |
| 2     | Older Alluvium      | 10000    | 5.068      | 0.01499   | 0.4277   | 63.34  | 991.9  | 4.821          | 0.0001314 | 0.1614  |
| 3     | Massive Carbonate 1 | 10000    | 24.07      | 0.1614    | 2.377    | 247.1  | 4280   | 4.424          | 6.417e-05 | 0.1074  |
| 4     | Massive Carbonate 2 | 10000    | 0.9773     | 0.005258  | 0.09286  | 10.11  | 293.8  | 4.747          | 3.558e-05 | 0.06777 |
| 5     | Fractured Granite   | 10000    | 0.05168    | 0.0002467 | 0.005391 | 0.4971 | 6.752  | 4.437          | 1.258e-05 | 0.02816 |

FAULT HEAD-JUMP DIAGNOSTICS

Mean absolute head difference across the blocked cell faces, compared with the mean head difference between ordinary neighbouring cells in the same layer. A contrast ratio near 1 would mean the barriers are invisible and the benchmark pointless; these values confirm a genuine internal discontinuity that a smooth neural representation cannot reproduce.

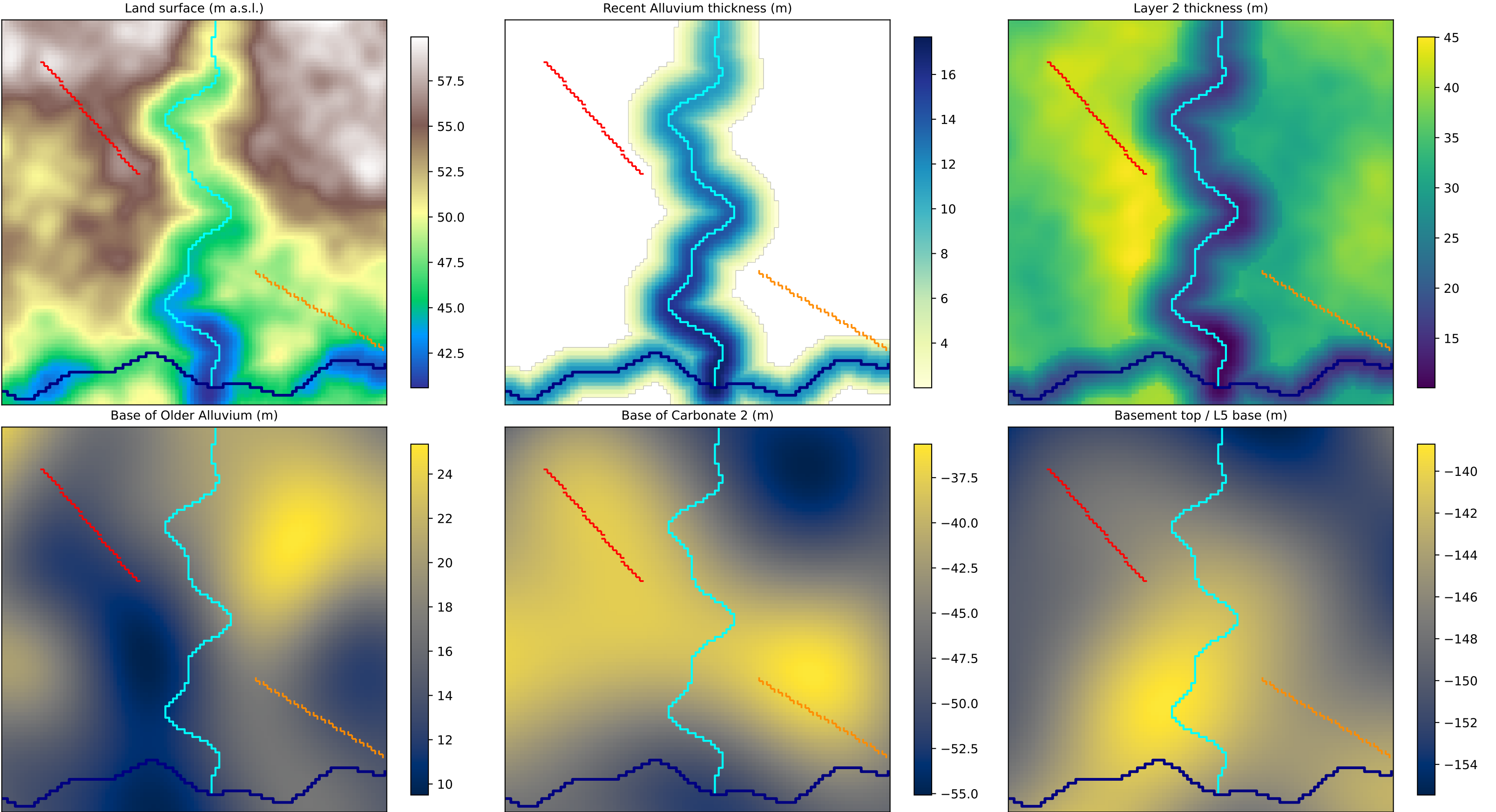
| layer | fault                | hydchr | n_faces | dh_mean_m | dh_max_m | background_dh_m | contrast_ratio |
|-------|----------------------|--------|---------|-----------|----------|-----------------|----------------|
| 1     | FAULT2_SEMIPERMEABLE | 0.001  | 7       | 2.312     | 3.147    | 0.184           | 12.56          |
| 2     | FAULT1_IMPERMEABLE   | 1e-08  | 55      | 0.6511    | 1.612    | 0.1184          | 5.502          |
| 2     | FAULT2_SEMIPERMEABLE | 0.001  | 54      | 1.521     | 3.158    | 0.1184          | 12.85          |
| 3     | FAULT1_IMPERMEABLE   | 1e-08  | 55      | 0.6091    | 1.732    | 0.1052          | 5.791          |
| 3     | FAULT2_SEMIPERMEABLE | 0.001  | 54      | 1.573     | 3.141    | 0.1052          | 14.96          |
| 4     | FAULT1_IMPERMEABLE   | 1e-08  | 55      | 0.6004    | 1.661    | 0.09927         | 6.048          |
| 4     | FAULT2_SEMIPERMEABLE | 0.001  | 54      | 1.464     | 3.023    | 0.09927         | 14.75          |
| 5     | FAULT1_IMPERMEABLE   | 1e-08  | 55      | 0.6145    | 1.602    | 0.09312         | 6.599          |
| 5     | FAULT2_SEMIPERMEABLE | 0.001  | 54      | 0.5405    | 1.185    | 0.09312         | 5.804          |

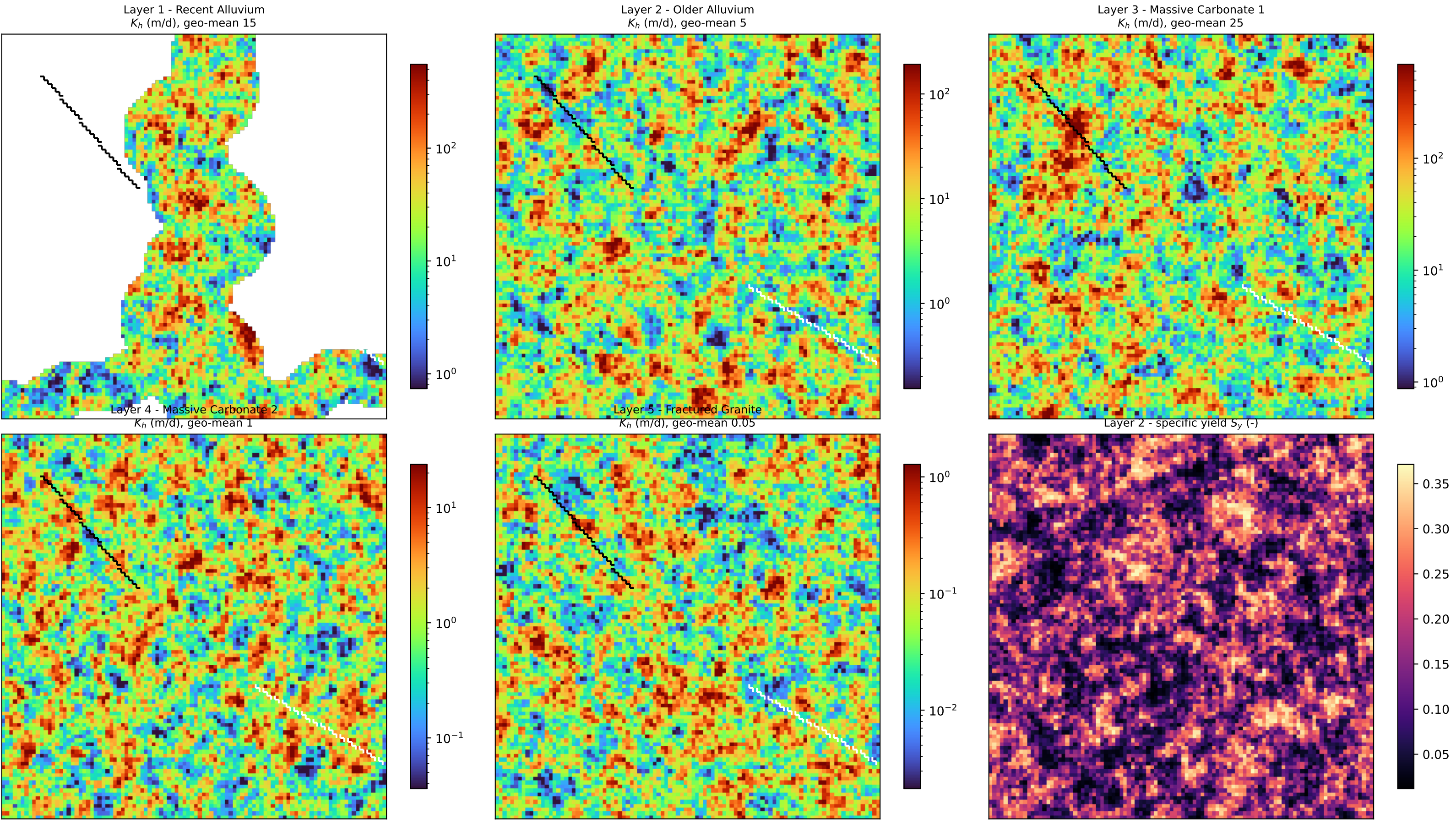
OBSERVATION POINTS

Four head observations, two on each side of the middle blocked face of each fault. The paired cells are 100 m apart but separated by the barrier, so their difference measures the discontinuity directly, at 2-hourly resolution.

|           |         |        |        |
|-----------|---------|--------|--------|
| F1_SIDE_A | layer 2 | row 25 | col 22 |
| F1_SIDE_B | layer 2 | row 25 | col 23 |
| F2_SIDE_A | layer 2 | row 75 | col 82 |
| F2_SIDE_B | layer 2 | row 75 | col 81 |

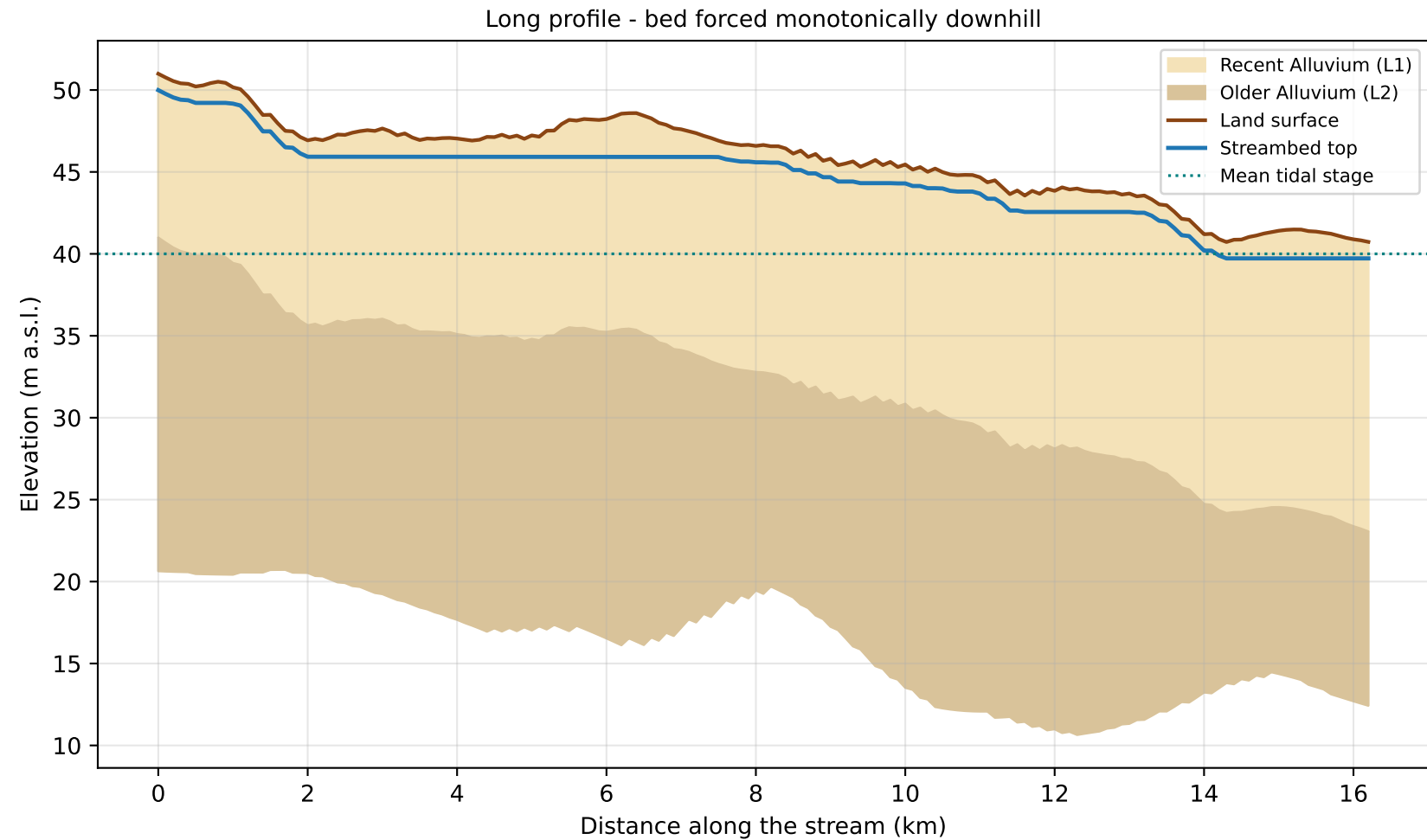
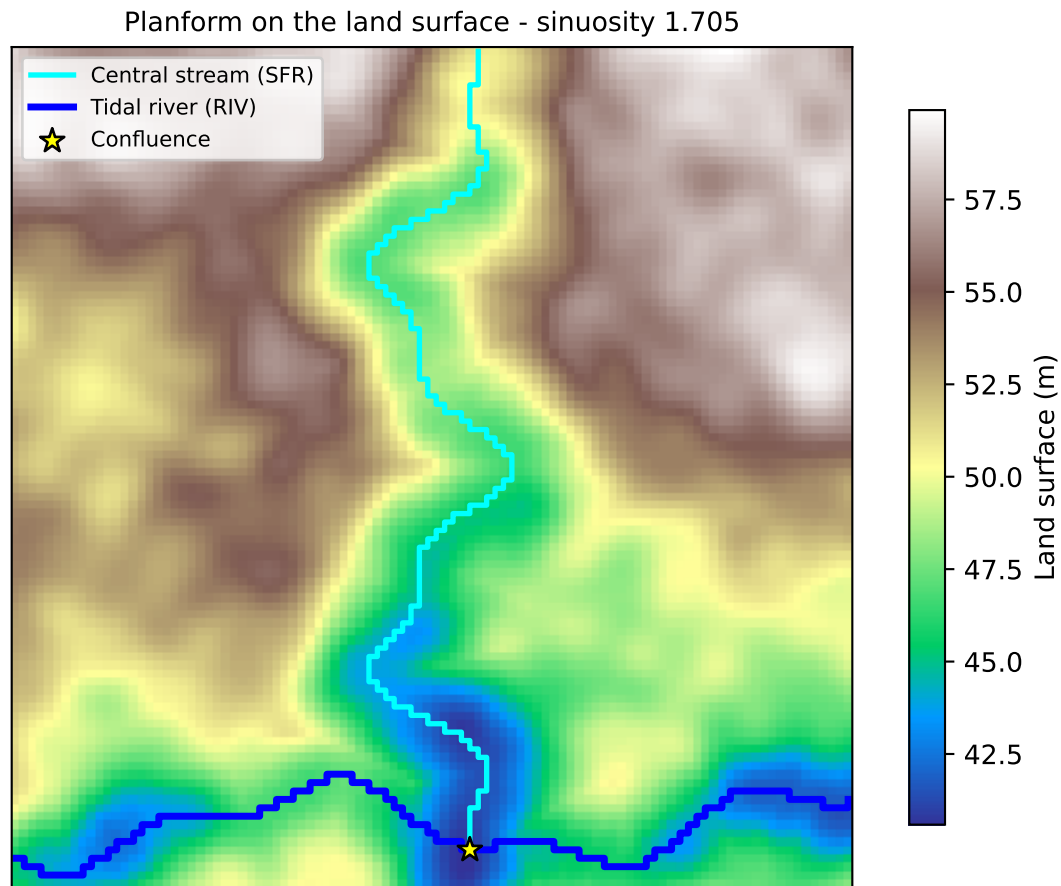
Topography and stratigraphic geometry - meandering stream (cyan), tidal river (blue), faults (red/orange)





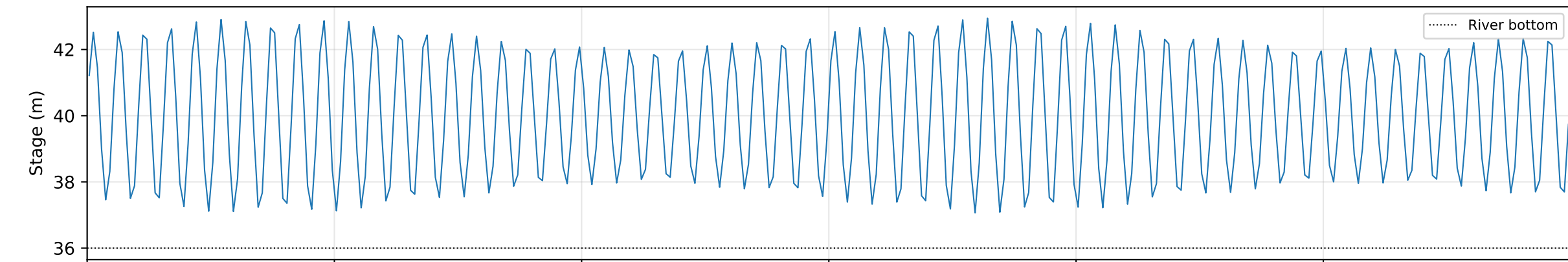


## Meandering central stream

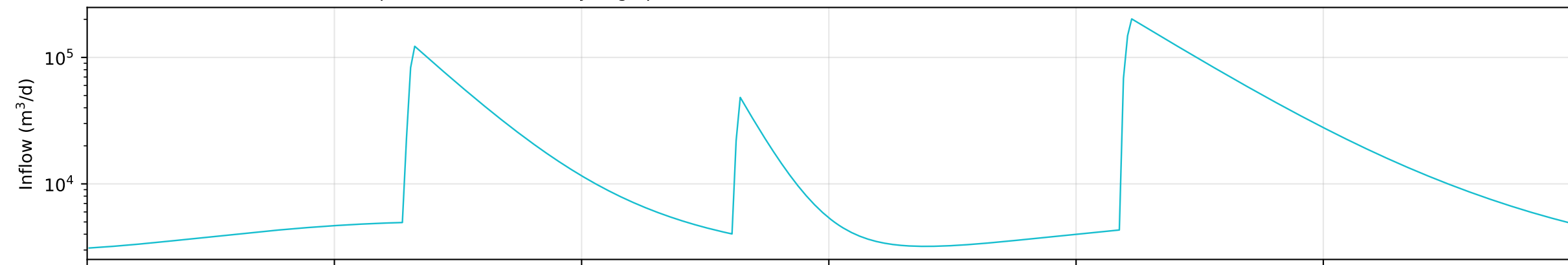


Transient forcing at 2-hour stress-period resolution

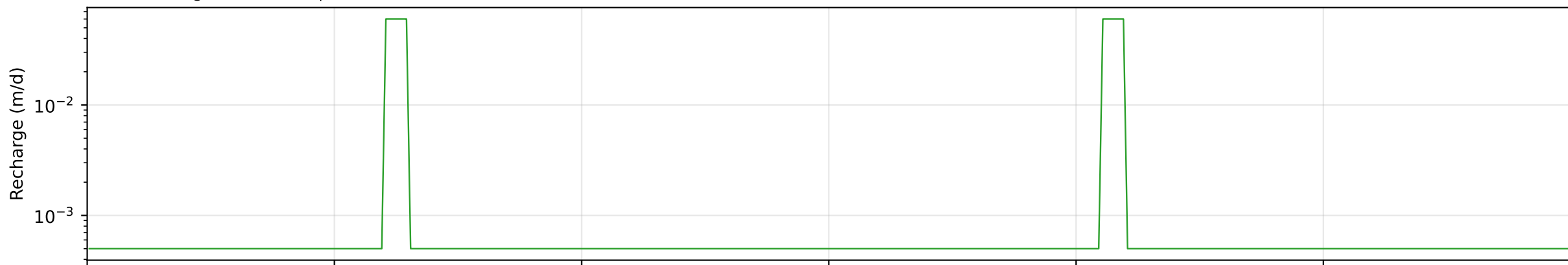
Tidal river stage - M2 semi-diurnal (12.4 h) with spring-neap envelope



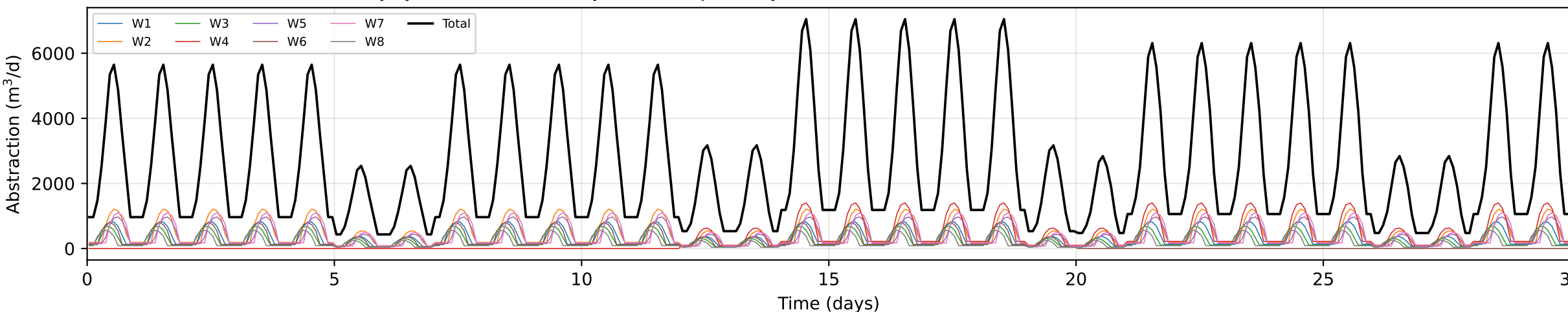
SFR head-reach inflow - baseflow plus three flash-flood hydrographs



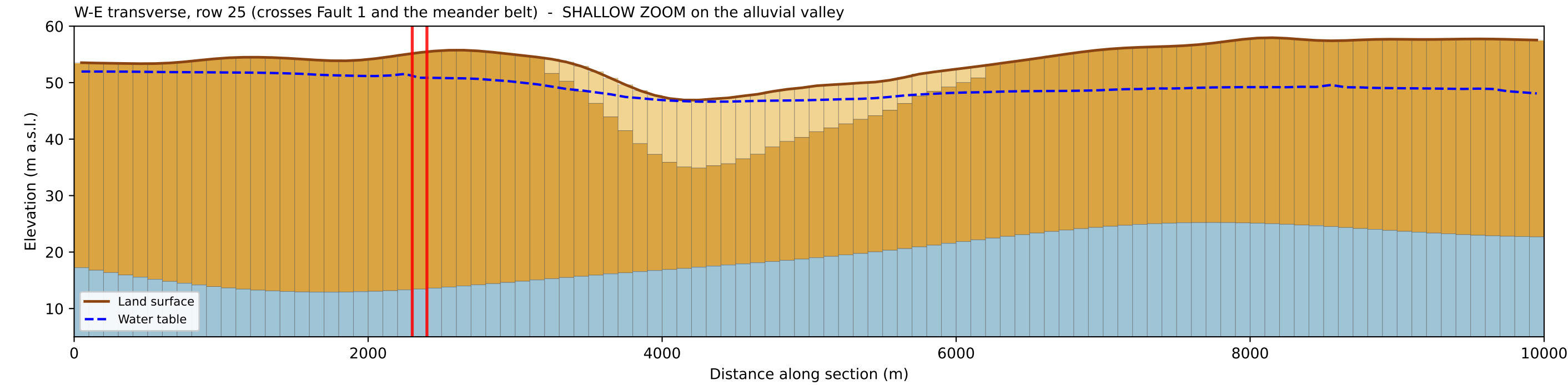
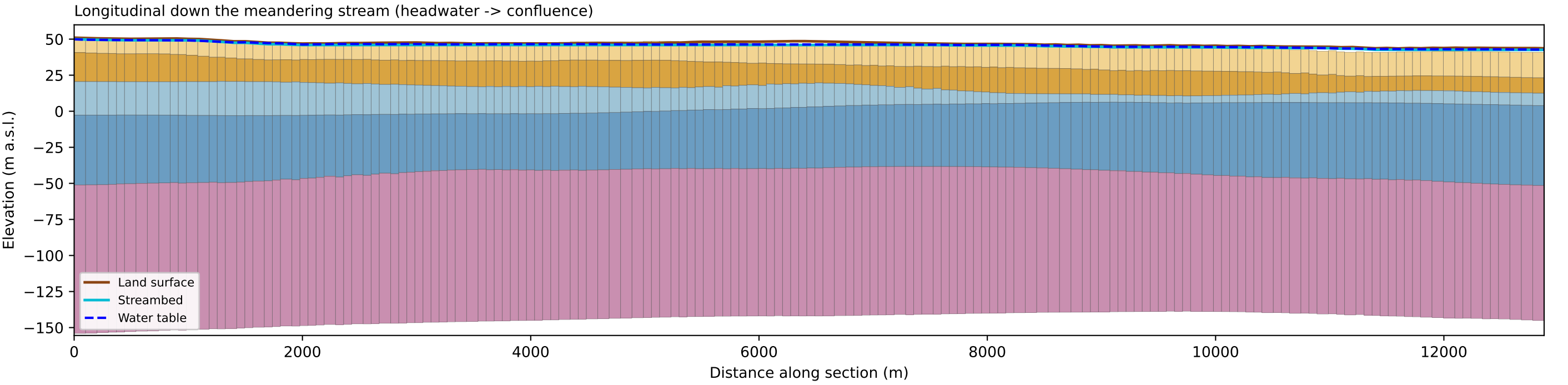
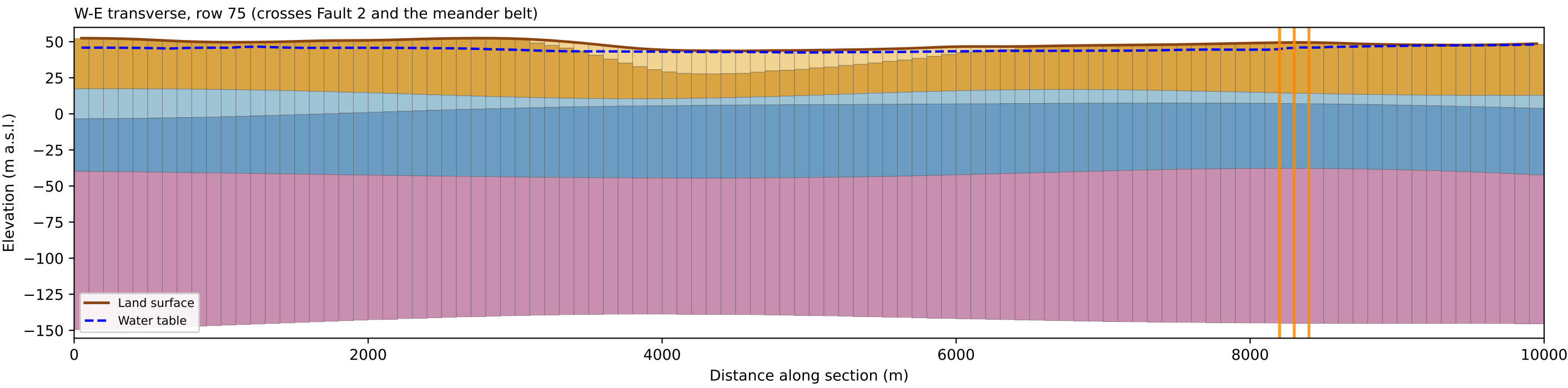
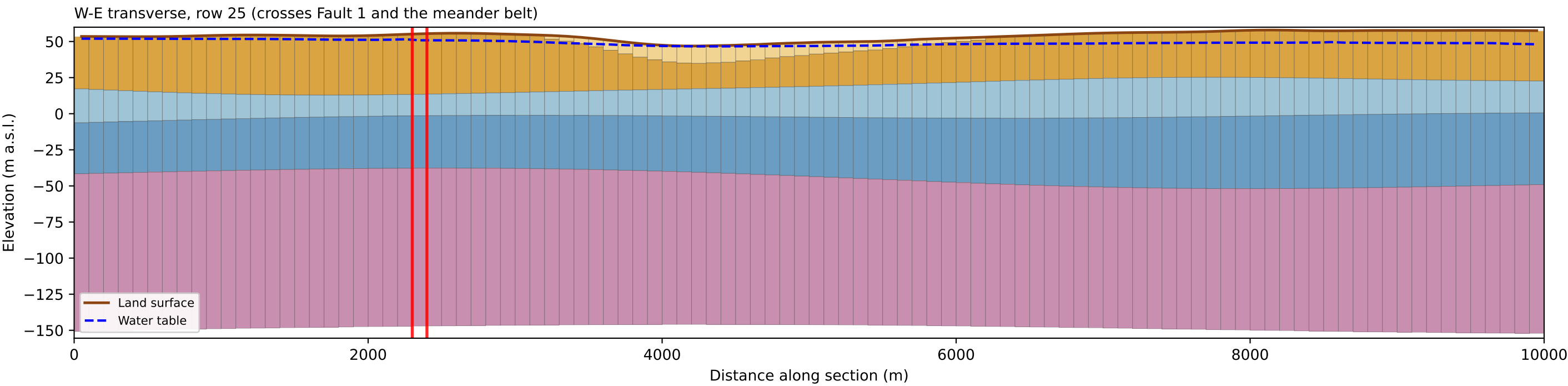
Areal recharge - base rate plus two intense 12-hour storms



Well abstraction - diurnal + weekly cycles, W4 starts at day 12, W6 stops at day 20

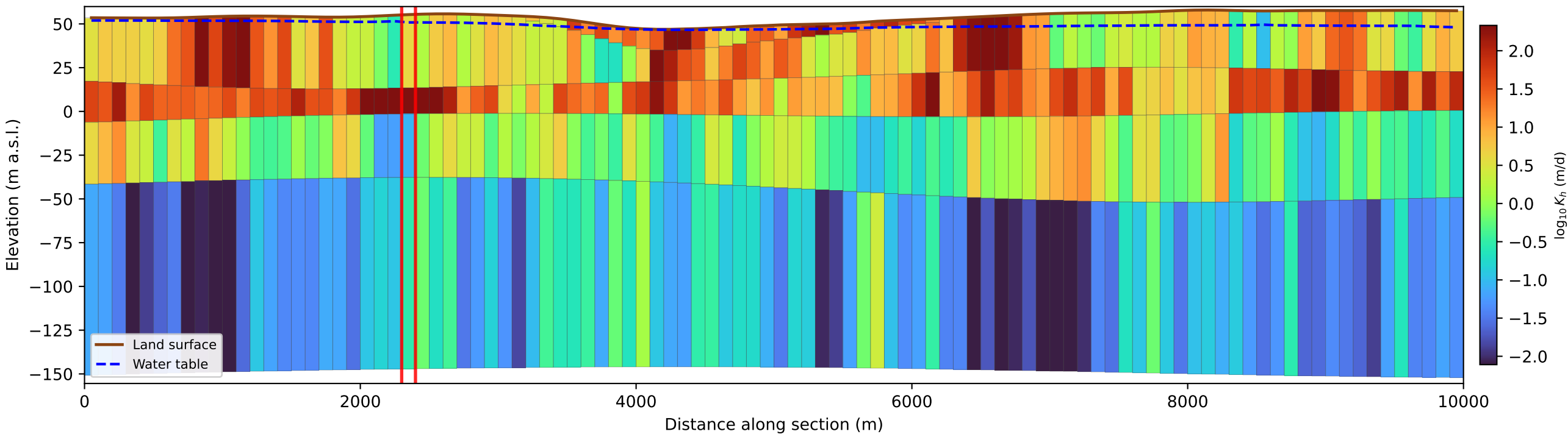


Geological cross-sections - stratigraphic units, undulating contacts and the alluvial pinch-out

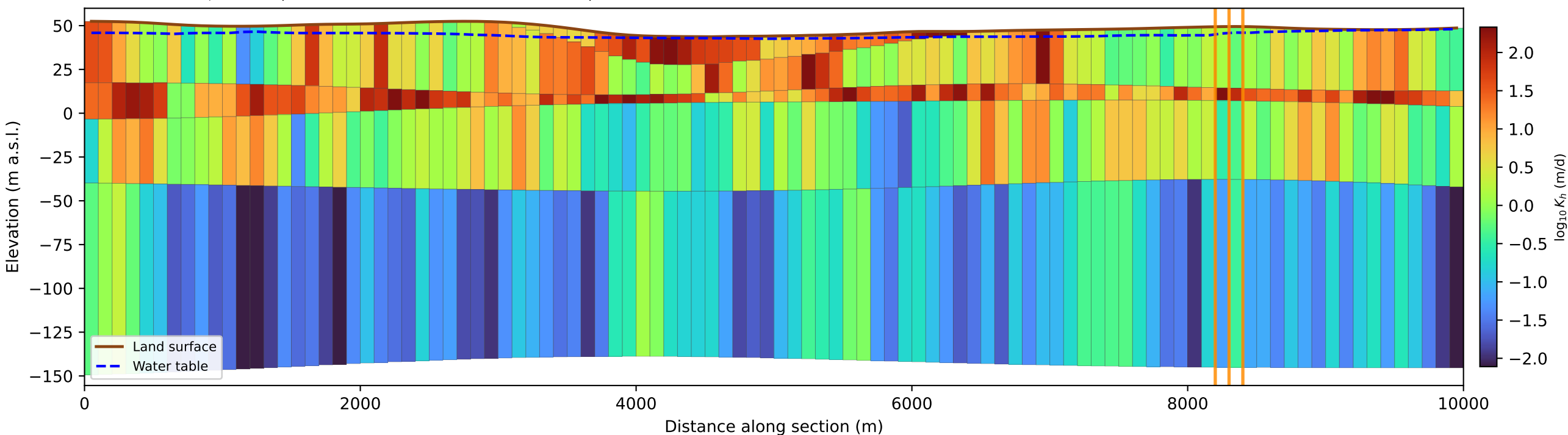


# Hydraulic conductivity in cross-section - two to three orders of magnitude of variation within every unit

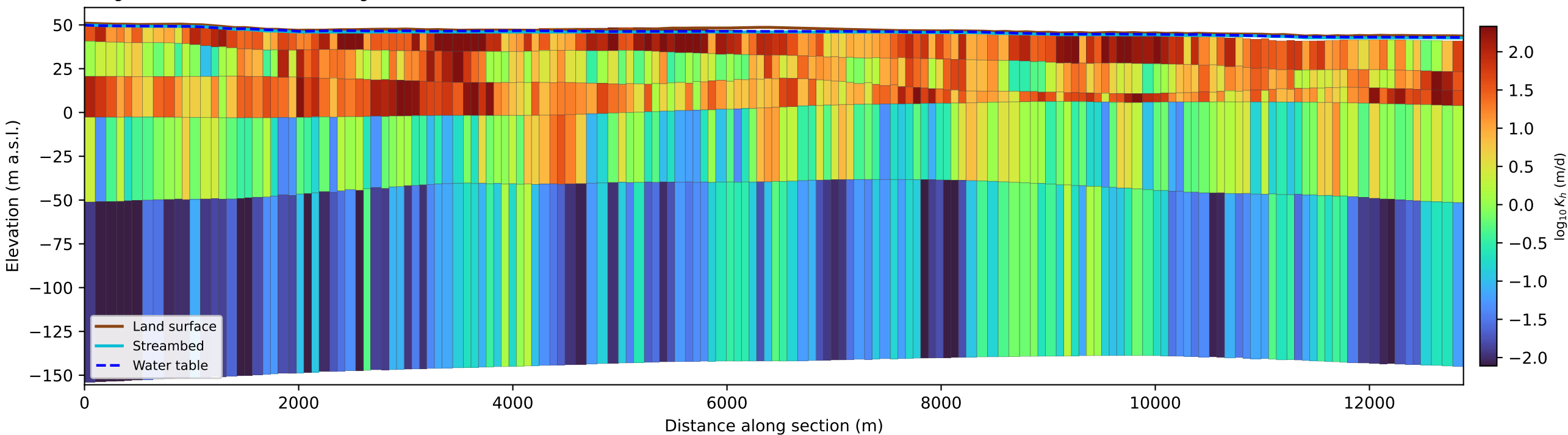
W-E transverse, row 25 (crosses Fault 1 and the meander belt)



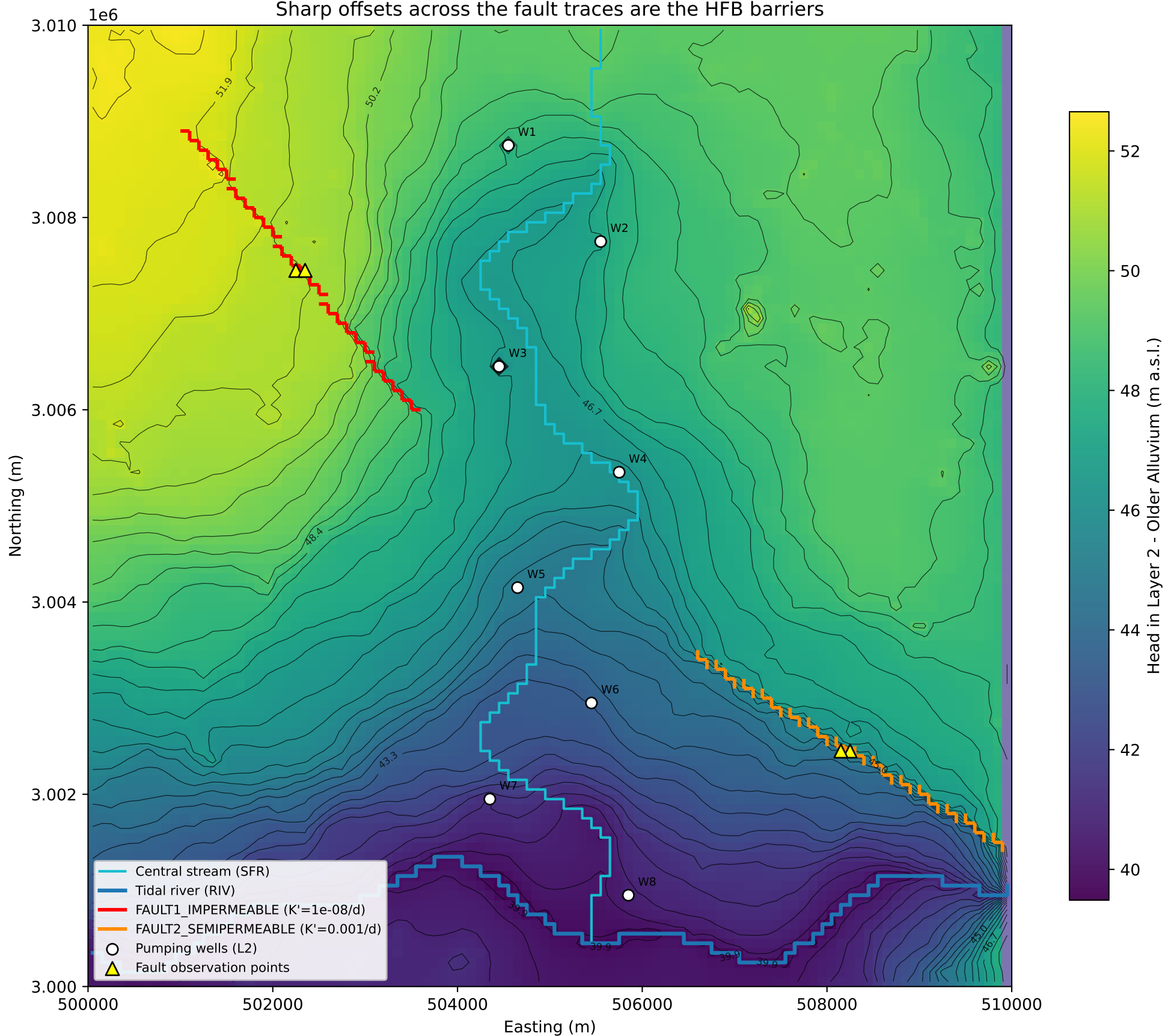
W-E transverse, row 75 (crosses Fault 2 and the meander belt)



Longitudinal down the meandering stream (headwater -> confluence)

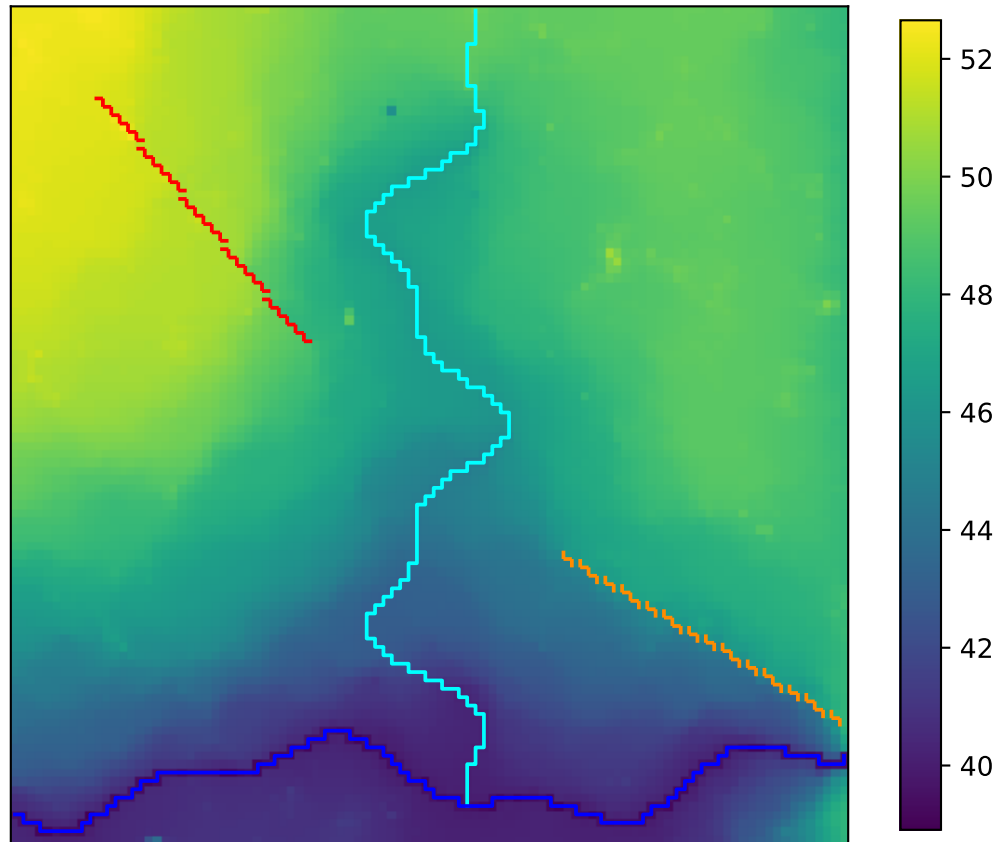


Layer 2 simulated head - stress period 180 (t = 15.000 d)  
 Sharp offsets across the fault traces are the HFB barriers

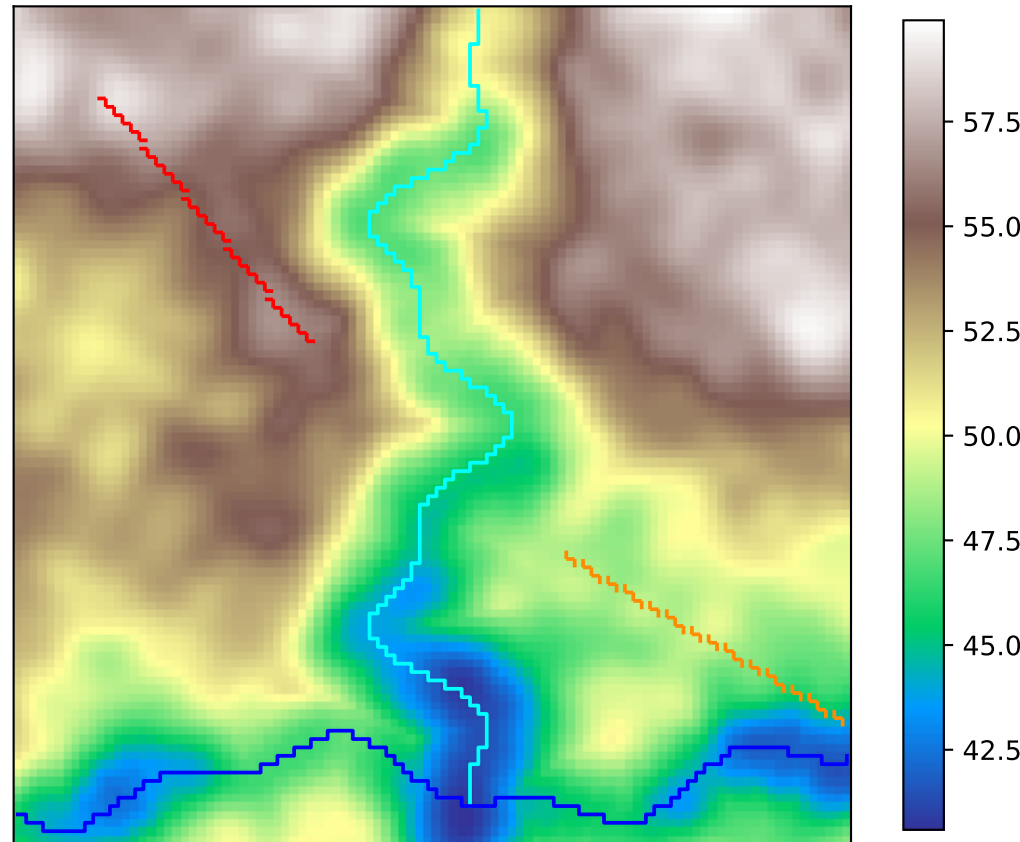


Water table at stress period 180. Negative depth = water table above land surface (see the module docstring)

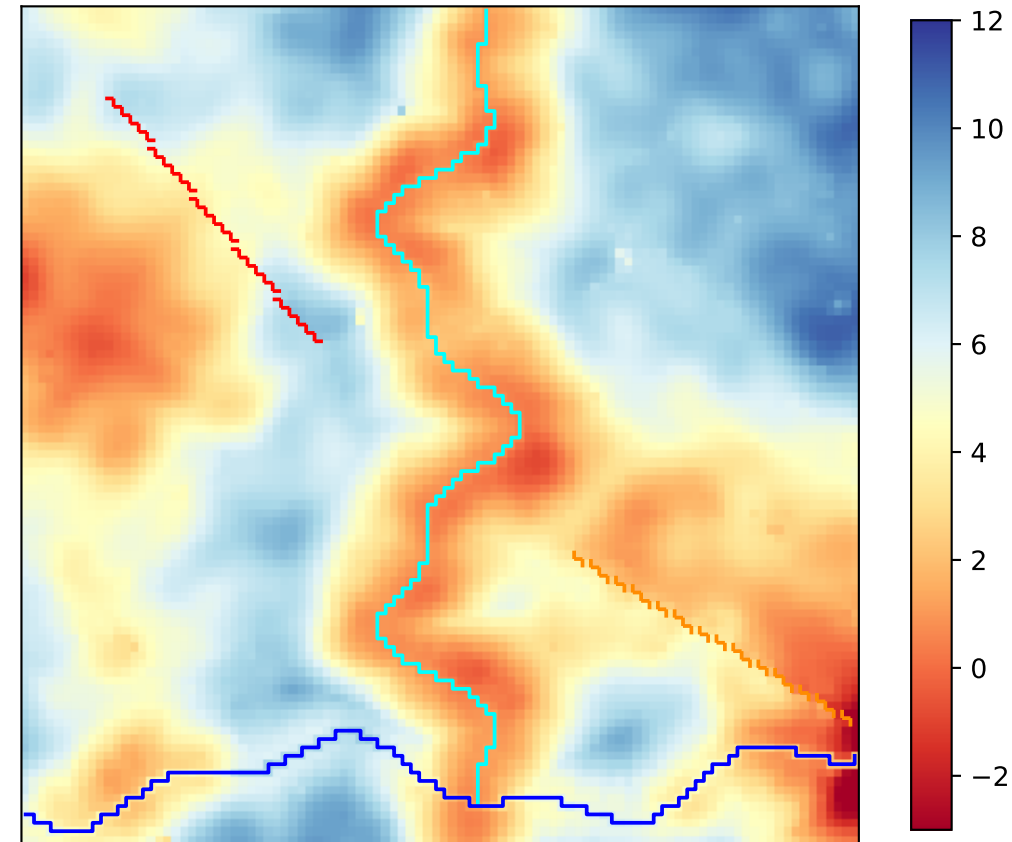
Water-table elevation (m a.s.l.)



Land surface (m a.s.l.)

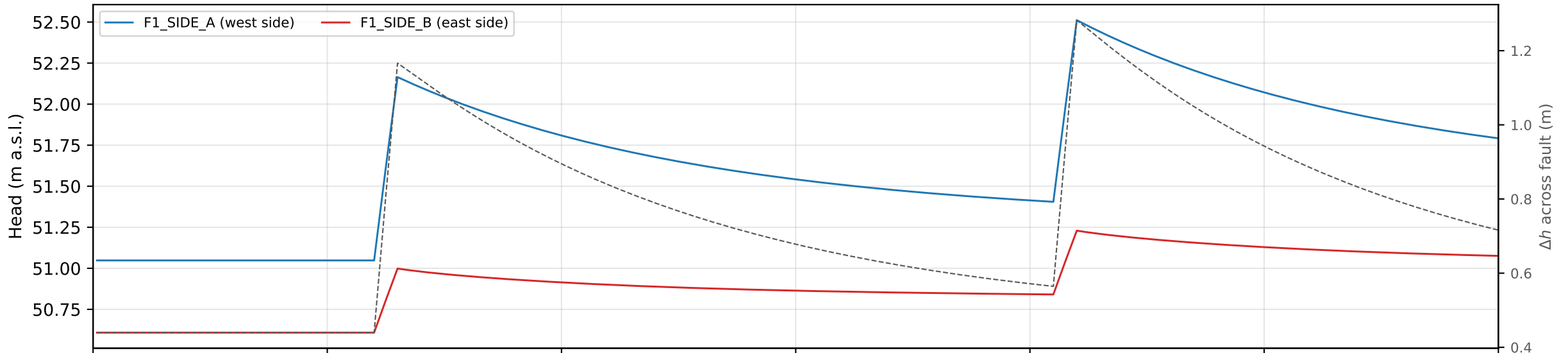


Depth to water (m below ground)

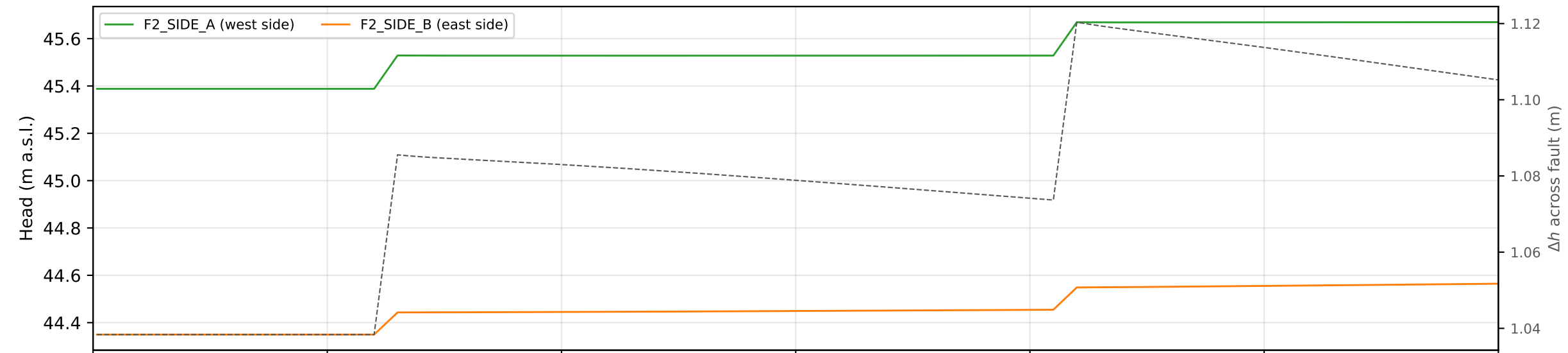


# Fault-straddling head hydrographs - 2-hourly resolution over 30 days

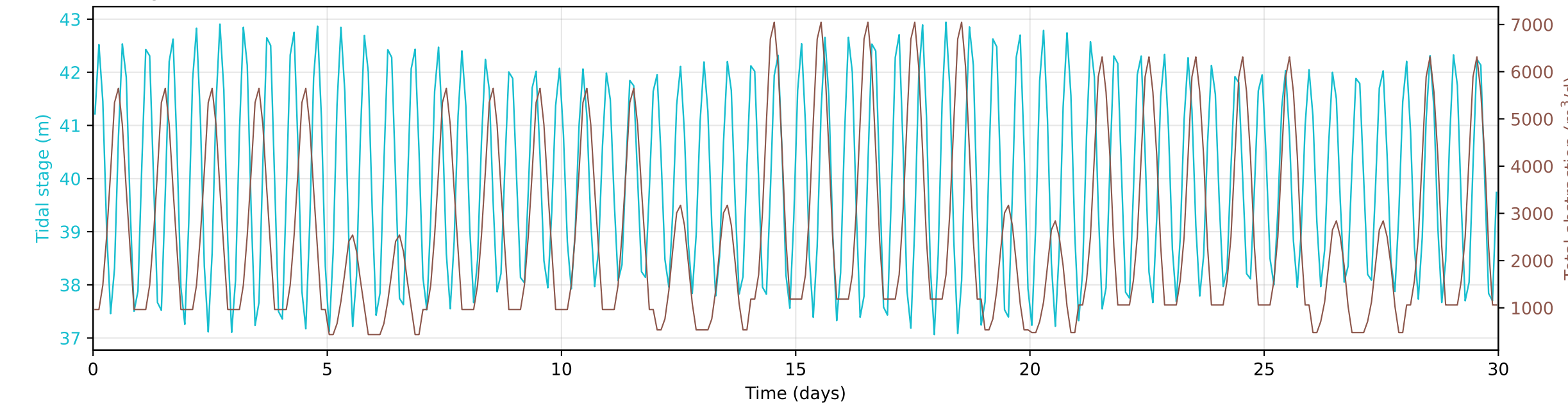
## Fault 1 - impermeable ( $\text{hydchr} = 1\text{e-}8 \text{ d}^{-1}$ )



## Fault 2 - semi-permeable ( $\text{hydchr} = 1\text{e-}3 \text{ d}^{-1}$ )



## Driving stresses (context)



# 4. Data inventory for surrogate training

Every field the benchmark uses or produces is written to georeferenced GIS files in EPSG:32644, so a surrogate can be trained, validated and inspected without ever reading a MODFLOW binary.

|         | files | size_mb |
|---------|-------|---------|
| folder  |       |         |
| arrays  | 1     | 48.6    |
| figures | 9     | 3.1     |
| gis     | 139   | 36.0    |
| report  | 1     | 1.9     |
| tables  | 8     | 0.1     |

## FOR THE FORWARD (SIMULATION) PROBLEM

inputs kh\_L\*, kz\_L\*, ss\_L\*, sy\_L\*, top\_L\*, botm\_L\*, thickness\_L\*, idomain\_L\*,  
icelltype\_L\*, land\_surface, alluvium\_thickness, recharge\_layer\_irch,  
strt\_L\* (the initial condition)  
forcing tables/forcing\_by\_stress\_period.csv - stage, stream inflow, recharge  
and all 8 well rates for each of the 360 stress periods  
targets head\_timeseries\_L\*.tif (multi-band, band n = stress period n)  
arrays/heads\_transient.npz (nper, nlay, nrow, ncol) float32

## FOR THE INVERSE (PARAMETER ESTIMATION) PROBLEM

unknowns kh\_L\*, ss\_L\*, sy\_L\* (and the fault hydchr values in faults\_trace.shp)  
data head\_timeseries\_L\*.tif, watertable\_timeseries.tif,  
tables/fault\_observations.csv (dense 2-hourly hydrographs),  
qx\_L\*/qy\_L\*/qz\_L\* specific discharge for Darcy-residual terms  
priors dist\_to\_stream, dist\_to\_river, dist\_to\_fault1, dist\_to\_fault2,  
dist\_to\_any\_fault, dist\_to\_nearest\_well, valley\_corridor\_mask -  
the covariates that carry the structural information

The PINN collocation inputs x\_coordinate.tif and y\_coordinate.tif give cell-centre coordinates in metres on the same grid, so (x, y, t) samples can be drawn directly without re-deriving the affine transform.

## RASTERS (128 GEOTIFFS)

| file                   | description                        |
|------------------------|------------------------------------|
| alluvium_thickness.tif | Recent Alluvium fill thickness (m) |
| botm_L1.tif            | Layer bottom elevation (m)         |
| botm_L2.tif            | Layer bottom elevation (m)         |



## 4. Data inventory for surrogate training (continued)

|                          |                                                            |
|--------------------------|------------------------------------------------------------|
| botm_L3.tif              | Layer bottom elevation (m)                                 |
| botm_L4.tif              | Layer bottom elevation (m)                                 |
| botm_L5.tif              | Layer bottom elevation (m)                                 |
| dist_to_any_fault.tif    | Euclidean distance covariate (m)                           |
| dist_to_fault1.tif       | Euclidean distance covariate (m)                           |
| dist_to_fault2.tif       | Euclidean distance covariate (m)                           |
| dist_to_nearest_well.tif | Euclidean distance covariate (m)                           |
| dist_to_river.tif        | Euclidean distance covariate (m)                           |
| dist_to_stream.tif       | Euclidean distance covariate (m)                           |
| drawdown_L1_sp180.tif    | Spin-up head minus snapshot head, i.e. drawdown (m)        |
| drawdown_L2_sp180.tif    | Spin-up head minus snapshot head, i.e. drawdown (m)        |
| drawdown_L3_sp180.tif    | Spin-up head minus snapshot head, i.e. drawdown (m)        |
| drawdown_L4_sp180.tif    | Spin-up head minus snapshot head, i.e. drawdown (m)        |
| drawdown_L5_sp180.tif    | Spin-up head minus snapshot head, i.e. drawdown (m)        |
| head_L1_sp180.tif        | Simulated head snapshot (m)                                |
| head_L2_sp180.tif        | Simulated head snapshot (m)                                |
| head_L3_sp180.tif        | Simulated head snapshot (m)                                |
| head_L4_sp180.tif        | Simulated head snapshot (m)                                |
| head_L5_sp180.tif        | Simulated head snapshot (m)                                |
| head_max_L1.tif          | Maximum simulated head over the 30 days (m)                |
| head_max_L2.tif          | Maximum simulated head over the 30 days (m)                |
| head_max_L3.tif          | Maximum simulated head over the 30 days (m)                |
| head_max_L4.tif          | Maximum simulated head over the 30 days (m)                |
| head_max_L5.tif          | Maximum simulated head over the 30 days (m)                |
| head_mean_L1.tif         | Time-mean simulated head (m)                               |
| head_mean_L2.tif         | Time-mean simulated head (m)                               |
| head_mean_L3.tif         | Time-mean simulated head (m)                               |
| head_mean_L4.tif         | Time-mean simulated head (m)                               |
| head_mean_L5.tif         | Time-mean simulated head (m)                               |
| head_min_L1.tif          | Minimum simulated head over the 30 days (m)                |
| head_min_L2.tif          | Minimum simulated head over the 30 days (m)                |
| head_min_L3.tif          | Minimum simulated head over the 30 days (m)                |
| head_min_L4.tif          | Minimum simulated head over the 30 days (m)                |
| head_min_L5.tif          | Minimum simulated head over the 30 days (m)                |
| head_range_L1.tif        | Max-minus-min simulated head, i.e. transient amplitude (m) |
| head_range_L2.tif        | Max-minus-min simulated head, i.e. transient amplitude (m) |
| head_range_L3.tif        | Max-minus-min simulated head, i.e. transient amplitude (m) |
| head_range_L4.tif        | Max-minus-min simulated head, i.e. transient amplitude (m) |
| head_range_L5.tif        | Max-minus-min simulated head, i.e. transient amplitude (m) |
| head_timeseries_L1.tif   | Multi-band head time stack; band n = stress period n (m)   |
| head_timeseries_L2.tif   | Multi-band head time stack; band n = stress period n (m)   |
| head_timeseries_L3.tif   | Multi-band head time stack; band n = stress period n (m)   |
| head_timeseries_L4.tif   | Multi-band head time stack; band n = stress period n (m)   |

## 4. Data inventory for surrogate training (continued)

|                         |                                                            |
|-------------------------|------------------------------------------------------------|
| head_timeseries_L5.tif  | Multi-band head time stack; band n = stress period n (m)   |
| icelltype_L1.tif        | Convertible flag (1 water table, 0 confined)               |
| icelltype_L2.tif        | Convertible flag (1 water table, 0 confined)               |
| icelltype_L3.tif        | Convertible flag (1 water table, 0 confined)               |
| icelltype_L4.tif        | Convertible flag (1 water table, 0 confined)               |
| icelltype_L5.tif        | Convertible flag (1 water table, 0 confined)               |
| idomain_L1.tif          | Active-cell mask (1 active, 0 pinched out)                 |
| idomain_L2.tif          | Active-cell mask (1 active, 0 pinched out)                 |
| idomain_L3.tif          | Active-cell mask (1 active, 0 pinched out)                 |
| idomain_L4.tif          | Active-cell mask (1 active, 0 pinched out)                 |
| idomain_L5.tif          | Active-cell mask (1 active, 0 pinched out)                 |
| kh_L1.tif               | Horizontal hydraulic conductivity - INVERSION TARGET (m/d) |
| kh_L2.tif               | Horizontal hydraulic conductivity - INVERSION TARGET (m/d) |
| kh_L3.tif               | Horizontal hydraulic conductivity - INVERSION TARGET (m/d) |
| kh_L4.tif               | Horizontal hydraulic conductivity - INVERSION TARGET (m/d) |
| kh_L5.tif               | Horizontal hydraulic conductivity - INVERSION TARGET (m/d) |
| kz_L1.tif               | Vertical hydraulic conductivity (m/d)                      |
| kz_L2.tif               | Vertical hydraulic conductivity (m/d)                      |
| kz_L3.tif               | Vertical hydraulic conductivity (m/d)                      |
| kz_L4.tif               | Vertical hydraulic conductivity (m/d)                      |
| kz_L5.tif               | Vertical hydraulic conductivity (m/d)                      |
| land_surface.tif        | Land surface elevation (m)                                 |
| log10_kh_L1.tif         | log10 horizontal hydraulic conductivity (log10 m/d)        |
| log10_kh_L2.tif         | log10 horizontal hydraulic conductivity (log10 m/d)        |
| log10_kh_L3.tif         | log10 horizontal hydraulic conductivity (log10 m/d)        |
| log10_kh_L4.tif         | log10 horizontal hydraulic conductivity (log10 m/d)        |
| log10_kh_L5.tif         | log10 horizontal hydraulic conductivity (log10 m/d)        |
| qx_L1_sp180.tif         | Specific discharge, x component (m/d)                      |
| qx_L2_sp180.tif         | Specific discharge, x component (m/d)                      |
| qx_L3_sp180.tif         | Specific discharge, x component (m/d)                      |
| qx_L4_sp180.tif         | Specific discharge, x component (m/d)                      |
| qx_L5_sp180.tif         | Specific discharge, x component (m/d)                      |
| qy_L1_sp180.tif         | Specific discharge, y component (m/d)                      |
| qy_L2_sp180.tif         | Specific discharge, y component (m/d)                      |
| qy_L3_sp180.tif         | Specific discharge, y component (m/d)                      |
| qy_L4_sp180.tif         | Specific discharge, y component (m/d)                      |
| qy_L5_sp180.tif         | Specific discharge, y component (m/d)                      |
| qz_L1_sp180.tif         | Specific discharge, z component (m/d)                      |
| qz_L2_sp180.tif         | Specific discharge, z component (m/d)                      |
| qz_L3_sp180.tif         | Specific discharge, z component (m/d)                      |
| qz_L4_sp180.tif         | Specific discharge, z component (m/d)                      |
| qz_L5_sp180.tif         | Specific discharge, z component (m/d)                      |
| recharge_layer_irch.tif | 1-based layer receiving areal recharge                     |

## 4. Data inventory for surrogate training (continued)

|                                     |                                                            |
|-------------------------------------|------------------------------------------------------------|
| specific_discharge_magnitude_L2.tif | Horizontal Darcy flux magnitude (m/d)                      |
| ss_L1.tif                           | Specific storage - INVERSION TARGET (1/m)                  |
| ss_L2.tif                           | Specific storage - INVERSION TARGET (1/m)                  |
| ss_L3.tif                           | Specific storage - INVERSION TARGET (1/m)                  |
| ss_L4.tif                           | Specific storage - INVERSION TARGET (1/m)                  |
| ss_L5.tif                           | Specific storage - INVERSION TARGET (1/m)                  |
| strt_L1.tif                         | Steady-state initial head used to start the transient r... |
| strt_L2.tif                         | Steady-state initial head used to start the transient r... |
| strt_L3.tif                         | Steady-state initial head used to start the transient r... |
| strt_L4.tif                         | Steady-state initial head used to start the transient r... |
| strt_L5.tif                         | Steady-state initial head used to start the transient r... |
| sy_L1.tif                           | Specific yield - INVERSION TARGET (-)                      |
| sy_L2.tif                           | Specific yield - INVERSION TARGET (-)                      |
| sy_L3.tif                           | Specific yield - INVERSION TARGET (-)                      |
| sy_L4.tif                           | Specific yield - INVERSION TARGET (-)                      |
| sy_L5.tif                           | Specific yield - INVERSION TARGET (-)                      |
| thickness_L1.tif                    | Layer geometric thickness (m)                              |
| thickness_L2.tif                    | Layer geometric thickness (m)                              |
| thickness_L3.tif                    | Layer geometric thickness (m)                              |
| thickness_L4.tif                    | Layer geometric thickness (m)                              |
| thickness_L5.tif                    | Layer geometric thickness (m)                              |
| top_L1.tif                          | Layer top elevation (m)                                    |
| top_L2.tif                          | Layer top elevation (m)                                    |
| top_L3.tif                          | Layer top elevation (m)                                    |
| top_L4.tif                          | Layer top elevation (m)                                    |
| top_L5.tif                          | Layer top elevation (m)                                    |
| transmissivity_L1.tif               | Kh x layer thickness (m2/d)                                |
| transmissivity_L2.tif               | Kh x layer thickness (m2/d)                                |
| transmissivity_L3.tif               | Kh x layer thickness (m2/d)                                |
| transmissivity_L4.tif               | Kh x layer thickness (m2/d)                                |
| transmissivity_L5.tif               | Kh x layer thickness (m2/d)                                |
| uppermost_active_layer.tif          | 1-based index of the highest active layer                  |
| valley_corridor_mask.tif            | Layer 1 presence mask (meander belt)                       |
| watertable_depth_sp180.tif          | Land surface minus water table (m below ground)            |
| watertable_range.tif                | Water-table fluctuation amplitude (m)                      |
| watertable_sp180.tif                | Water-table elevation (m)                                  |
| watertable_timeseries.tif           | Multi-band water-table time stack (m)                      |
| x_coordinate.tif                    | Cell-centre easting (m) - PINN collocation input           |
| y_coordinate.tif                    | Cell-centre northing (m) - PINN collocation input          |

## 5. Data inventory (continued)

### SHAPEFILES (11)

| file                         | description                                     |
|------------------------------|-------------------------------------------------|
| bc_ghb_cells.shp             | Eastern regional-inflow GHB cells               |
| bc_riv_cells.shp             | Tidal-river RIV cells                           |
| bc_riv_centerline.shp        | Meandering tidal-river centreline               |
| bc_sfr_centerline.shp        | Meandering stream centreline                    |
| bc_sfr_reaches.shp           | SFR reaches with streambed geometry             |
| bc_wells.shp                 | Production wells with scheduling metadata       |
| domain_outline.shp           | Model domain rectangle                          |
| faults_hfb_faces.shp         | Every blocked cell face used by the HFB package |
| faults_trace.shp             | Idealised straight fault traces with hydchr     |
| obs_points.shp               | Fault-straddling head observation points        |
| valley_alluvial_corridor.shp | Polygonised meander-belt alluvium               |

### TABLES AND ARRAYS

| file                           | size_kb | description                                       |
|--------------------------------|---------|---------------------------------------------------|
| MANIFEST.csv                   | 10.0    |                                                   |
| fault_contrast_diagnostics.csv | 1.0     | Head jump across faults vs background gradient    |
| fault_observations.csv         | 20.8    | Observed head hydrographs at the 4 obs points     |
| forcing_by_stress_period.csv   | 72.8    | Every boundary forcing, per stress period         |
| observation_points.csv         | 0.2     | Observation point coordinates                     |
| property_statistics.csv        | 1.0     | Per-layer hydraulic property statistics           |
| stream_long_profile.csv        | 10.6    | SFR streambed long profile                        |
| wells.csv                      | 0.6     | Well locations and pumping metadata               |
| heads_transient.npz            | 49802.9 | Full head tensor (nper, nlay, nrow, ncol) float32 |

### REPRODUCING THIS DATASET

```
python scripts/build_mf6_hetero_benchmark.py --seed 20260805
```

Change --seed for an independent realisation of the same conceptual model: the geometry, the channel planforms and the faults are deterministic, while the K, Ss, Sy and geological-surface fields are not. --nper shortens the run for smoke tests and --no-run stops after writing the model and the static exports.